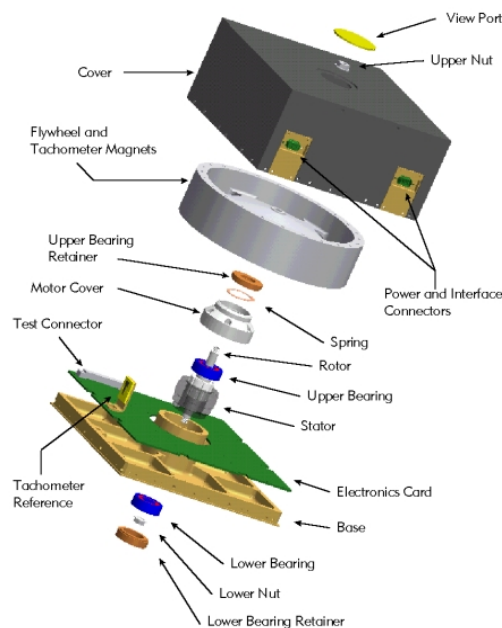


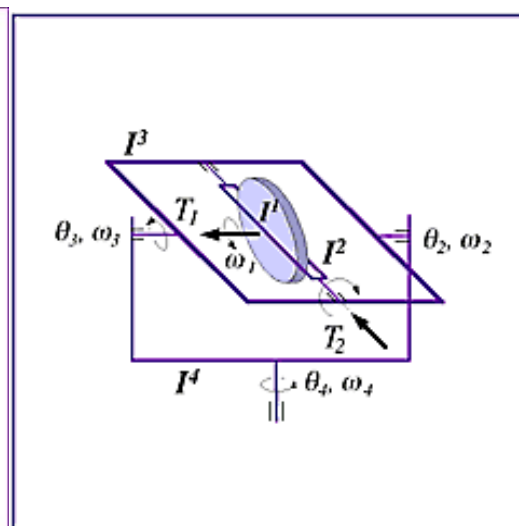
Chapter 5. Momentum Exchange Systems

Momentum exchange devices can absorb angular momentum disturbances, and can be used to change the attitude of spacecraft.

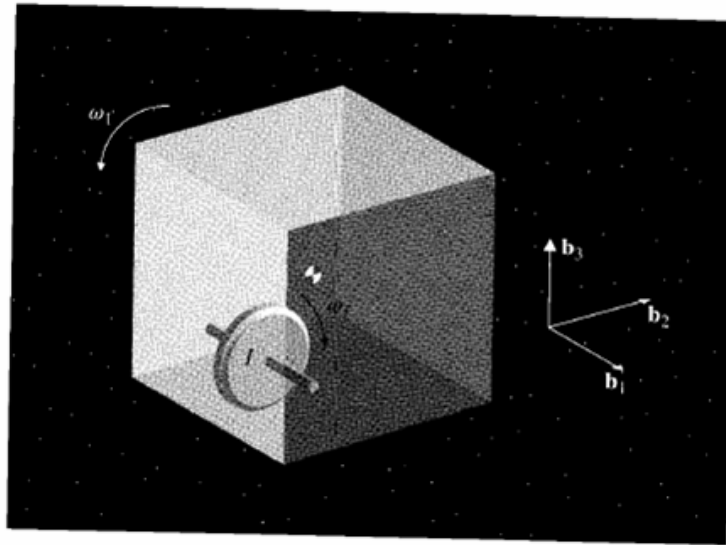
Reaction/Momentum wheels: They have fixed spin direction. They absorb angular momentum disturbances by changing their spin speed.



Control Momentum Gyros (CMGs): They have constant spin speed and they have one or two gimbals. They absorb angular momentum disturbances by gimbal rotation (i.e. changing spin direction.)



Reaction Wheel Systems



Principle: We begin with the satellite stationary with respect to inertial frame, and the flywheel not rotating.

Attitude control is performed without using thrusters (force).

→ No external torques (No external force is applied.)

→ angular momentum is conserved ($\dot{\vec{H}} = 0$)

$\vec{H} = 0$; the total angular momentum $\vec{H}_{tot}$ of this system is then initially zero.

The moment of inertia of Spacecraft : A

The moment of inertia of the flywheel : I

(typically $I \ll A$, A does not include I)

Consider angle-axis motion of about $\hat{b}_1$ as an electric motor to rotate the wheel at angular rate $\omega_{flywheel}$ compared to the spacecraft.

The equal and opposite reaction torque acts on the spacecraft causes the satellite to rotate at rate ω_1 with respect to inertial space.

→ Angular momentum conservation

magnitude of $\vec{H}_{tot}$: $H_{tot} = I(\omega_{flywheel} - \omega_1) - A\omega_1 = 0$ (with respect to inertial frame)

$$\rightarrow \omega_1 = \frac{I\omega_{flywheel}}{A + I}$$

Other case for initial conditions : $\omega_1(0) \neq 0, \omega_{flywheel}(0) = 0$ (i.e. $\vec{H}(t=0) \neq 0$)

→ the wheel is spun up using an electric motor.

$$H_o = (I + A)\omega_1(0) \text{ in inertial space,}$$

$$H_f = I(\omega_{flywheel} - \omega_1) - A\omega_1$$

Note: $\omega_{flywheel}$ with respect to the spacecraft

⇒ This system is not excited by any disturbances, the total angular momentum is conserved in inertial space.

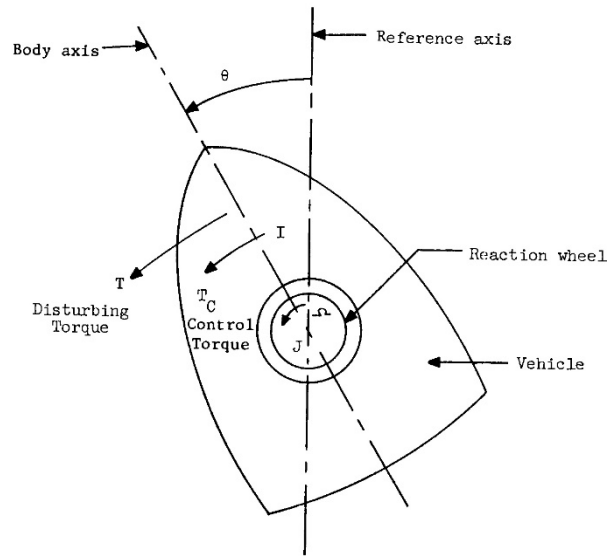
$$H_o = H_f \rightarrow I\omega_1(0) + A\omega_1(0) = I(\omega_{flywheel}(t) - \omega_1(t)) - A\omega_1(t)$$

We can control $\omega_1(t)$ by selecting proper values of $\omega_{flywheel}$

$$\rightarrow \text{If we want to } \omega_1(t)=0, \text{ then } \omega_{flywheel} = \frac{(A+I)\omega_1(0)}{I}$$

; The direction of $\omega_{flywheel}$ will be the same as $\omega_1(0)$.

Example:



Angular momentum of wheel with respect to the inertial frame is

$$H_w = J(\Omega + \dot{\theta}) \quad (1)$$

Angular momentum of spacecraft with respect to the inertial frame is

$$H_{s/c} = I\dot{\theta} \quad (2)$$

For a single degree of freedom planar motion, the equation of motion is

$$T = \frac{d}{dt}(H_w + H_{s/c}) \quad (3)$$

$$= (I + J)\ddot{\theta} + J\dot{\Omega} \quad (4)$$

Taking Laplace transform

$$(I + J)s^2\Theta(s) + Js\Omega(s) = T(s) \quad (5)$$

$$\dot{\theta}(0)=0, \quad \ddot{\theta}(0)=0, \quad \Omega(0)=0$$

For spacecraft

$$Is^2\Theta(s) = T(s) - T_c(s) \quad (6)$$

$T_c(s)$; wheel control torque

For wheel only, (Equation(5) - Equation(6))

$$Js(\Omega(s) + s\Theta(s)) = T_c(s) \quad (7)$$

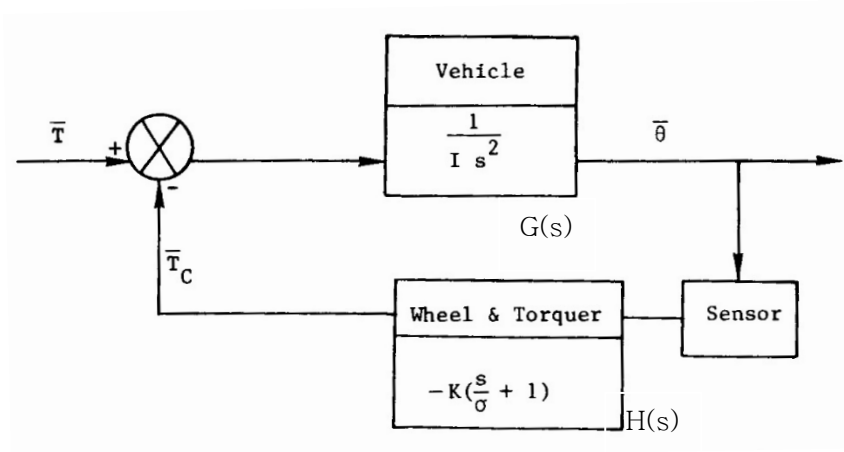


Fig 5.2 Reaction wheel control system

Let the gain

$$K \equiv \frac{I}{\tau^2}, \quad \tau; \text{time constant}$$

$$G(s) = \frac{1}{Is^2}$$

and

$$H(s) = K \left(\frac{s}{\sigma} + 1 \right) \quad (\text{from: } K_R s + K_A, K_R: \text{rate gain}, K_A: \text{position gain})$$

The closed-loop transfer function

$$\frac{\Theta(s)}{T(s)} = \frac{G}{1+GH} = \frac{1}{Is^2 + K \left(\frac{s}{\sigma} + 1 \right)}$$

The poles of the equation

$$s_{1,2} = \frac{-K/\sigma \pm \sqrt{(K/\sigma)^2 - 4KI}}{2I}$$

For critical damping,

$$\left(\frac{K}{\sigma} \right)^2 = 4KI$$

Then,

$$s_{1,2} = -2\sigma : \text{the two roots are negative and equal.}$$

Time constant

$$\tau^2 = \frac{I}{K} = \frac{I}{4I\sigma^2}, \quad \tau = \frac{1}{2\sigma}$$

Then,

$$\begin{aligned} \frac{\Theta(s)}{T(s)} &= \frac{1}{K \left(\frac{I}{K}s^2 + \frac{s}{\sigma} + 1 \right)} = \frac{1}{K(\tau^2 s^2 + 2\tau s + 1)} = \frac{\tau^2}{I(\tau s + 1)^2} \\ &= \frac{1}{I \left(s + \frac{1}{\tau} \right)^2} \end{aligned}$$

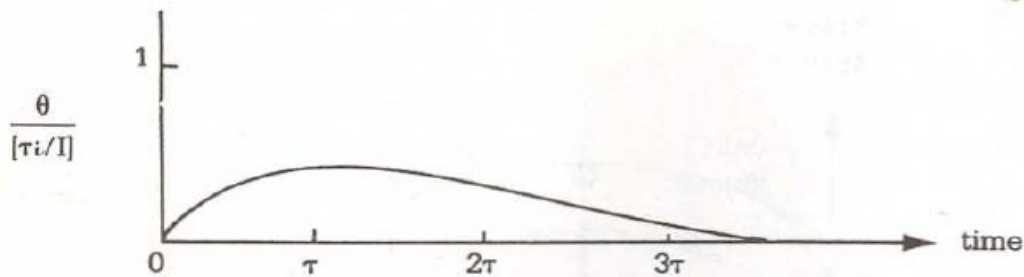


Figure 5.4 Response to an impulse torque of magnitude i .

For a torque impulse of magnitude, i

$$\Theta(s) = \frac{i}{I \left(s + \frac{1}{\tau} \right)^2}$$

$$\theta(t) = \frac{i\tau}{I} \cdot \frac{t}{\tau} e^{-t/\tau} \quad : \text{the attitude error is decreased with respect to time}$$

Maximum system response occurs at $t = \tau$

$$\theta_{\max}(t) = \frac{i\tau}{I} \cdot \frac{1}{e}$$

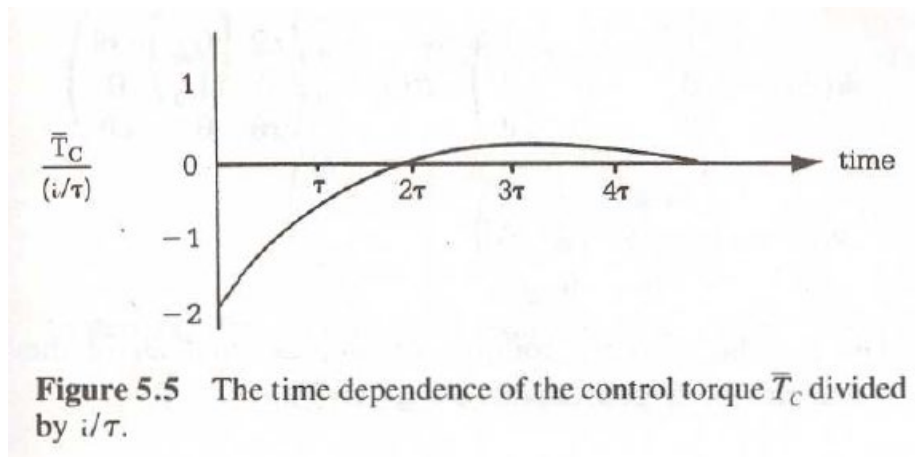
From figure 5.2, the control torque is

$$\begin{aligned} T_c(s) &= K \left(\frac{s}{\sigma} + 1 \right) \Theta, \quad (K = I/\tau^2, \sigma = 1/2\tau, \Theta = \tau^2 T / (I(\tau s + 1)^2)) \\ &= \frac{I}{\tau^2} (2\tau s + 1) \frac{\tau^2 T}{I(\tau s + 1)^2} \\ &= \frac{(2\tau s + 1)T}{(\tau s + 1)^2} \end{aligned}$$

For $T = i$ (impulse torque)

$$T_c(s) = \frac{(2\tau s + 1)i}{(\tau s + 1)^2}$$

$$T_c(t) = \frac{2i}{\tau} \left(1 - \frac{t}{2\tau} \right) e^{-t/\tau}$$



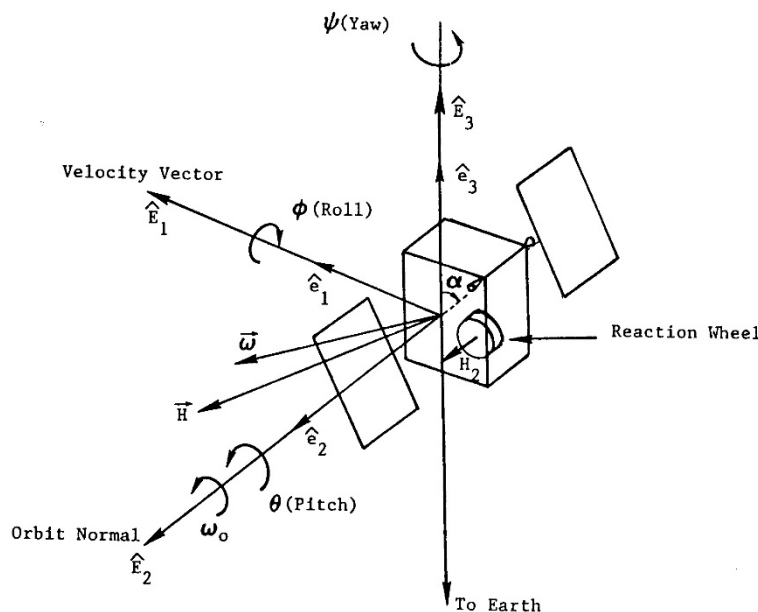
Momentum Bias System

The momentum bias system uses a momentum wheel to provide inertial stiffness in two axes, and control of wheel speed provides control in the third axis.

A single reaction wheel (RW) is aligned along the pitch axis of the spacecraft which is oriented along the normal to the orbital plane. The reaction wheel assembly is an inertia wheel that controls the pointing of the spacecraft about the pitch during the normal mode of operation. Acceleration or deceleration of the RW produces corrective torques on the spacecraft to maintain the pointing accuracy of the spacecraft.

If the spacecraft rotates away from its nominal position about its yaw axis, then the bias momentum of the RW will couple with the spacecraft's orbit rate providing gyroscope torques to return the spacecraft to its proper orientation about the yaw axis.

Pitch momentum wheel



e_1, e_2, e_3 ; body frame

E_1, E_2, E_3 ; orbit frame

<Equations of motion>

$$\vec{T} = \dot{\vec{H}} + \vec{\omega} \times \vec{H}$$

$$\vec{H} = I_1 \omega_1 \hat{e}_1 + I_2 \omega_2 \hat{e}_2 + I_3 \omega_3 \hat{e}_3 + h_2 \hat{e}_2$$

where h_2 is the RW angular momentum along the $\hat{e}_2$ axis

$$\vec{\omega} = (\omega_1, \omega_2, \omega_3)$$

$$\begin{aligned} \vec{\omega} \times \vec{H} &= \begin{vmatrix} \hat{e}_1 & \hat{e}_2 & \hat{e}_3 \\ \omega_1 & \omega_2 & \omega_3 \\ I_1 \omega_1 & I_2 \omega_2 + h_2 & I_3 \omega_3 \end{vmatrix} \\ &= [(I_3 - I_2) \omega_2 \omega_3 - h_2 \omega_3] \hat{e}_1 \\ &\quad + (I_1 - I_3) \omega_1 \omega_3 \hat{e}_2 \\ &\quad + [(I_2 - I_1) \omega_1 \omega_2 + h_2 \omega_1] \hat{e}_3 \end{aligned}$$

$$\dot{\vec{H}} = I_1 \dot{\omega}_1 \hat{e}_1 + I_2 \dot{\omega}_2 \hat{e}_2 + I_3 \dot{\omega}_3 \hat{e}_3 + \dot{h}_2 \hat{e}_2$$

Note :

$$\vec{H} = \vec{H}_s + \vec{h}_2$$

s means spacecraft

$$\vec{\omega} \times \vec{H} = (\vec{\omega} \times \vec{H}_s) + (\vec{\omega} \times \vec{h}_2)$$

$$= -h_2 \omega_3 \hat{e}_1 + h_2 \omega_1 \hat{e}_3 \quad \text{roll - yaw stiffness}$$

Equation of motion becomes

$$T_1 = I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_2 \omega_3 - h_2 \omega_3$$

$$T_2 = I_2 \dot{\omega}_2 + \dot{h}_2 + (I_1 - I_3) \omega_1 \omega_3 \quad (1)$$

$$T_3 = I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2 + h_2 \omega_1$$

$$\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = R_3(\psi)R_2(\theta)R_1(\phi) \begin{bmatrix} E_1 \\ E_2 \\ E_3 \end{bmatrix}$$

The angular velocity components in the body coordinates are given by

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + R(\psi) \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + R(\psi)R(\theta) \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} + R(\psi)R(\theta)R(\phi) \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix}$$

$$R(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{pmatrix}, \quad R(\theta) = \begin{pmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{pmatrix}$$

$$R(\psi) = \begin{pmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Small angle approximation

$$\cos(\text{angle}) \approx 1, \quad \sin(\text{angle}) \approx \text{angle}$$

$$R(\psi) \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} \approx \begin{pmatrix} 1 & \psi & 0 \\ -\psi & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} = \begin{bmatrix} \psi \dot{\theta} \\ \dot{\theta} \\ 0 \end{bmatrix} \approx \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix}, \quad \text{because } \psi \text{ is small and } \theta \text{ is small.}$$

$$R(\psi)R(\theta) \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} \approx \begin{bmatrix} 1 & \Psi & 0 \\ -\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -\theta \\ 0 & 1 & 0 \\ \theta & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & \psi & -\theta \\ -\psi & 1 & \psi \theta \\ \theta & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \dot{\phi} \\ -\psi \dot{\phi} \\ \theta \dot{\phi} \end{bmatrix} \approx \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \approx \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} + R(\psi)R(\theta)R(\phi) \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix}$$

$$\approx \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} + \omega_0 \begin{bmatrix} \psi \\ 1 \\ -\phi \end{bmatrix}$$

$$\begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} \approx \begin{bmatrix} \ddot{\phi} + \omega_0 \dot{\psi} \\ \ddot{\theta} \\ \ddot{\psi} - \omega_0 \dot{\phi} \end{bmatrix}$$

$$\omega_2 \omega_3 \approx (\dot{\theta} + \omega_0)(\dot{\psi} - \omega_0 \phi) \approx \omega_0 \dot{\psi} - \omega_0^2 \phi$$

$$\omega_1 \omega_3 \approx (\dot{\phi} + \omega_0 \psi)(\dot{\psi} - \omega_0 \phi) \approx 0$$

$$\omega_1 \omega_2 \approx (\dot{\phi} + \omega_0 \psi)(\dot{\theta} + \omega_0) \approx \omega_0 \dot{\phi} + \omega_0^2 \psi$$

Equation (1) becomes

$$T_1 = I_1 \ddot{\phi} + [\omega_0^2 (I_2 - I_3) + \omega_0 h_2] \phi + [\omega_0 (I_1 - I_2 + I_3) - h_2] \dot{\psi}$$

$$T_2 = I_2 \ddot{\theta} + \dot{h}_2 \tag{2}$$

$$T_3 = I_3 \ddot{\psi} + [\omega_0^2 (I_2 - I_1) + \omega_0 h_2] \psi - [\omega_0 (I_1 - I_2 + I_3) - h_2] \dot{\phi}$$

The roll(ϕ)and yaw(ψ) motion are uncoupled from the pitch motion (θ).

<Active pitch control>

The pitch attitude is controlled with the torque capabilities of the momentum wheel, $\dot{h}_2$. The momentum of the wheel is set initially at some nominal bias value and may change because of external disturbances acting about the pitch axis.

The equation of motion about pitch axis

$$I_2 \ddot{\theta} + \dot{h}_2 = T_2 \quad (1)$$

The equation indicated that no natural damping (no $\dot{\theta}$ term) is available. The control torque can be provided through the control $\dot{h}_2$.

The control law is

$$\dot{h}_2 = K_p (\theta + \tau_p \dot{\theta}) \quad : \text{contains a damping term} \quad (2)$$

where K_p and τ_p are the pitch gain and the pitch time constant.

Then, pitch motion is damped second order system.

The pitch equation becomes

$$I_2 \ddot{\theta} + K_p \tau_p \dot{\theta} + K_p \theta = T_2 \quad (3)$$

The associated transfer function is

$$\frac{\Theta(s)}{T_2(s)} = \frac{1}{I_2 s^2 + K_p \tau_p s + K_p} \quad (4)$$

Eq. (3) yields the pitch natural frequency and damping factor

$$\ddot{\theta} + \frac{K_p \tau_p}{I_2} \dot{\theta} + \frac{K_p}{I_2} \theta = \frac{T_2}{I_2} \equiv F_2$$

$$\ddot{\theta} + 2\frac{\tau_p}{2} \frac{K_p}{I_2} \dot{\theta} + \omega_p^2 \theta = F_2$$

$$\text{Let} \quad \omega_p^2 = \frac{K_p}{I_2}, \quad \omega_p = \sqrt{\frac{K_p}{I_2}} \quad (5)$$

$$\text{Let} \quad \varepsilon_p = \frac{\tau_p}{2} \sqrt{\frac{K_p}{I_2}} = \frac{\tau_p}{2} \omega_p \quad (6)$$

With Eqs. (5) and (6), Eq. (4) becomes

$$\Theta(s) = \frac{T_2(s)}{I_2(s^2 + 2\varepsilon_p \omega_p s + \omega_p^2)} \quad (7)$$

The performance of a control system is generally expressed in terms of acceptable steady-state error for disturbances. The steady-state error is(for step input => $T_2(s) = T_2/s$)

$\therefore$ from final – value theorem

$$\lim_{s \rightarrow 0} s \Theta(s) = s \frac{T_2}{I_2 s^2 + K_p \tau_p s + K_p} \frac{1}{s} = \frac{T_2}{K_p}$$

$$\theta_{ss} = \frac{T_2}{K_p} \text{ radians} \quad (8)$$

Where K_p is in Nm (because T_2 is in Nm)

Using eq. (8), the value of the pitch gain is chosen so that the steady-state error is within a given maximum. Then, we can determine the time constant τ_p to achieve a desired damping factor using eq. (6). The value of τ_p should be significantly smaller than the orbital period. τ_p is in second.

$$\therefore K_p = \frac{T_2}{\theta_{ss}} \quad (9)$$

(θ_{ss} : given or desired, T_2 : known or assume a worst case)

$$\therefore \tau_p = \frac{2\varepsilon_p}{\sqrt{\frac{K_p}{I_2}}} \quad (10)$$

(I_2 : given (N·m·sec²), K_p : calculated, ε_p : given for a desired damping)

Note:

$$\ddot{x} + 2\beta\dot{x} + \gamma^2 x = 0$$

i) underdamping : $\gamma^2 > \beta^2$

ii) overdamping : $\gamma^2 < \beta^2$

iii) critical damping : $\gamma^2 = \beta^2$

<Roll- Yaw Control>

The angular momentum of wheel (h_2) is practically large enough to neglect the terms of $\omega_0^2(I_2 - I_3)$, $\omega_0(I_1 - I_2 + I_3)$, $\omega_0^2(I_2 - I_1)$. The errors introduced in coefficients of ϕ and ψ are less than 1%.

The linearized equation of roll-yaw motion become

$$I_1\ddot{\phi} - h_2\dot{\psi} + \omega_0 h_2 \phi = T \cos \alpha + T_{s1} \quad \text{roll} \quad (1a)$$

$$I_3\ddot{\psi} + h_2\dot{\phi} + \omega_0 h_2 \psi = T \sin \alpha + T_{s3} \quad \text{yaw} \quad (1b)$$

Thruster direction is parallel to pitch axis.

α : angle between thruster direction and yaw axis

T: the magnitude of the torque delivered by the offset thrusters.

T_{s1} , T_{s2} : the external torque (disturbance) on roll and yaw axis.

h_2 provides damping terms $(\dot{\psi}, \dot{\phi})$ related with roll and yaw.

The control configuration based on roll angle only is popular because yaw sensing is expensive. Yaw error can be controlled by controlling the roll error because of roll-yaw coupling. The control torque command equation is

$$T = -K(\phi + \tau \dot{\phi}) \quad (2)$$

From eq. (1) and eq. (2), we obtain

$$I_1 \ddot{\phi} + K\tau \cos \alpha \dot{\phi} + (\omega_0 h_2 + K \cos \alpha) \phi - h_2 \dot{\psi} = T_{s1} \quad (3a)$$

$$I_3 \ddot{\psi} + \omega_0 h_2 \psi + (K\tau \sin \alpha + h_2) \dot{\phi} + K \sin \alpha \phi = T_{s3} \quad (3b)$$

The steady state solutions can be obtained from eqs. (3)

($\ddot{\phi} \rightarrow 0$, $\dot{\phi} \rightarrow 0$, $\dot{\psi} \rightarrow 0$, $\phi \rightarrow \phi_{ss}$ or final-value theorem).

$$(\omega_0 h_2 + K \cos \alpha) \phi_{ss} = T_{s1}$$

$$\phi_{ss} = \frac{T_{s1}}{\omega_0 h_2 + K \cos \alpha} \quad (4)$$

And $\ddot{\psi} \rightarrow 0$, $\dot{\psi} \rightarrow 0$, $\dot{\phi} \rightarrow 0$, $\psi \rightarrow \psi_{ss}$

$$\omega_0 h_2 \psi_{ss} + K \sin \alpha \phi_{ss} = T_{s3}$$

$$\psi_{ss} = \frac{T_{s3} - K \sin \alpha \phi_{ss}}{\omega_0 h_2} \quad (5)$$

It can be seen that ϕ_{ss} and ψ_{ss} are inversely proportional to wheel momentum h_2 .
Note : Although there is no control ($K=0$) for roll and yaw error, ϕ_{ss} and ψ_{ss} are bounded due to h_2 . This means that “RW couples with the spacecraft’s orbit rate providing gyroscope torques”.

For zero roll error $\phi_{ss} = 0$ and $T_{s1} = 0$, the yaw angle dynamics are determined from

equation (1a) which becomes

$$\begin{aligned} h_2 \dot{\psi} &= -T \cos \alpha \\ \rightarrow T &= -\frac{h_2 \dot{\psi}}{\cos \alpha} \end{aligned}$$

From equation (1b), we have

$$\begin{aligned} I_3 \ddot{\psi} + \omega_0 h_2 \psi &= T \sin \alpha + T_{s3} \\ &= -\left(\frac{h_2 \dot{\psi}}{\cos \alpha} \right) \sin \alpha + T_{s3} \end{aligned} \quad (6)$$

Therefore,

$$I_3 \ddot{\psi} + h_2 \dot{\psi} \tan \alpha + \omega_0 h_2 \psi = T_{s3} \quad (7)$$

Equation (7) shows that h_2 and α determine the damping characteristics of yaw (ψ) motions as well as roll(ϕ) motion through in terms of eqs. (1).

Steady-state solution $\ddot{\psi} \rightarrow 0$, $\dot{\psi} \rightarrow 0$, $\psi \rightarrow \psi_{ss}$

$$\psi_{ss} = \frac{T_{s3}}{\omega_0 h_2} \quad (8)$$

Thus, ψ_{ss} is directly proportional to the steady-state yaw disturbance and inversely proportional to h_2 (wheel momentum).

We can determine the size of h_2 from eq. (8).

$$h_2 = \frac{T_{s3}}{\omega_0 \psi_{ss}} \quad (\psi_{ss} : \text{given}, T_{s3} : \text{observed or worst case}) \quad (9)$$

From equation (4) we can determine K

$$K = \frac{T_{s1}}{\cos \alpha \phi_{ss}} - \frac{\omega_0 h_2}{\cos \alpha} \quad (10)$$

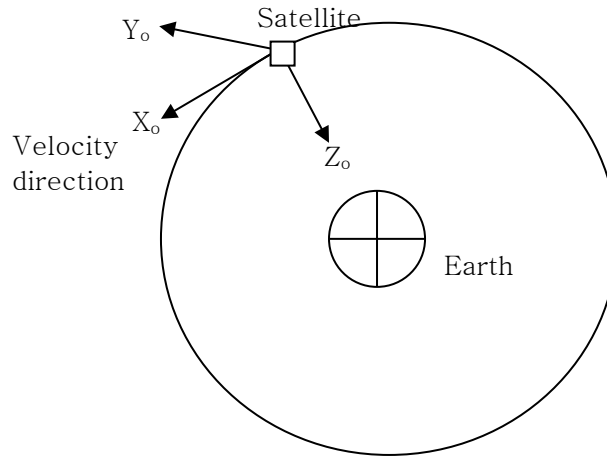
Note: when $\phi_{ss} \neq 0$, h_2 and K can be obtained using eqs. (4) and (5).

τ can be determined from eq. (3a) letting $\dot{\psi} = 0$, when desired damping ratio is given.

Three Axis Reaction Wheel Control

A three axis reaction wheel system can be considered as a combination of three independent roll, pitch and yaw control systems.

The system requires a reaction wheel with zero nominal angular momentum and an attitude sensor for each axis. Each axis is controlled by varying the speed of reaction wheel in response to the attitude error.



$$\vec{T} = \dot{\vec{H}} + \vec{\omega} \times \vec{H}$$

$$\text{where } \vec{H} = I_1 \omega_1 \hat{e}_1 + I_2 \omega_2 \hat{e}_2 + I_3 \omega_3 \hat{e}_3 + h_1 \hat{e}_1 + h_2 \hat{e}_2 + h_3 \hat{e}_3$$

$$\begin{aligned} T_1 &= \dot{h}_1 + I_1(\ddot{\phi} + \omega_0 \dot{\psi}) + h_3(\dot{\theta} + \omega_0) - h_2(\dot{\psi} - \omega_0 \phi) + (I_3 - I_2)\omega_0(\dot{\psi} - \omega_0 \phi) \\ &= I_1 \ddot{\phi} + \dot{h}_1 + [\omega_0^2(I_2 - I_3)\omega_0 h_2] \phi + [-h_2 + \omega_0(I_1 - I_2 + I_3)] \dot{\psi} + h_3 \dot{\theta} + \omega_0 h_3 \end{aligned}$$

$$\begin{aligned} T_2 &= \dot{h}_2 + I_2 \ddot{\theta} + h_1(\dot{\psi} - \omega_0 \phi) - h_3(\dot{\phi} + \omega_0 \psi) \\ &= I_2 \ddot{\theta} + \dot{h}_2 + h_1 \dot{\psi} - h_1 \omega_0 \phi - h_3 \dot{\phi} - h_3 \omega_0 \psi \end{aligned}$$

$$\begin{aligned} T_3 &= \dot{h}_3 + I_3(\ddot{\psi} - \omega_0 \dot{\phi}) + h_2(\dot{\phi} + \omega_0 \psi) - h_1(\dot{\theta} + \omega_0) + (I_2 - I_1)\omega_0(\dot{\phi} + \omega_0 \psi) \\ &= I_3 \ddot{\psi} + \dot{h}_3 + [\omega_0^2(I_2 - I_1)\omega_0 h_2] \psi + [h_2 - \omega_0(I_1 - I_2 + I_3)] \dot{\phi} - h_1 \dot{\theta} - \omega_0 h_1 \end{aligned}$$

Although the equations above are coupled, the coupling terms are small because h_1 , h_2 , h_3 and ω_0 are small.

If the coupling terms are neglected, the equations of motion about the roll, pitch, yaw axes becomes independent, and hence they can be controlled independently.

$$T_1 \approx I_1 \ddot{\phi} + \dot{h}_1$$

$$T_2 \approx I_2 \ddot{\theta} + \dot{h}_2$$

$$T_3 \approx I_3 \ddot{\psi} + \dot{h}_3$$

The control torques are applied by letting the rates of change of the angular momentum of reaction wheels have

$$\dot{h}_1 = K_\phi (\phi + \tau_\phi \dot{\phi})$$

$$\dot{h}_2 = K_\theta (\theta + \tau_\theta \dot{\theta})$$

$$\dot{h}_3 = K_\psi (\psi + \tau_\psi \dot{\psi})$$

The control dynamics of each axis very similar to the pitch axis control dynamics of the momentum bias system.

Control Moment Gyro Systems

Ref.: Howard D. Curtis, "Orbital Mechanics for Engineering Students,"
Elsevier Aerospace Engineering Series, 2005, pp. 526-530.

Momentum wheels that are free to pivot on one or more gimbals are called control moment gyros. Figure 10.24 illustrates a double-gimbaled control moment gyro. These gyros spin at several thousand rpm. The motor-driven speed of the flywheel is constant, and moments are exerted on the vehicle when torquers (electric motors) tilt the wheel about a gimbal axis. The torque direction is normal to the gimbal axis. To simplify the analysis of high-rpm gyros, we can assume that the angular momentum is directed totally along the spin axis. That is, in calculating the angular momentum $\mathbf{H}_{G_w}^{(w)}$ of a momentum wheel about its center of mass, we use the formula

$$\{\mathbf{H}_{G_w}^{(w)}\} = [\mathbf{I}_{G_w}^{(w)}]\{\boldsymbol{\omega}^{(w)}\}$$

where $\boldsymbol{\omega}^{(w)}$ is the absolute angular velocity of the spinning flywheel, which may be written

$$\boldsymbol{\omega}^{(w)} = \boldsymbol{\omega}^{(v)} + \boldsymbol{\omega}_p^{(w)} + \boldsymbol{\omega}_n^{(w)} + \boldsymbol{\omega}_s^{(w)}$$

$\boldsymbol{\omega}^{(v)}$ is the angular velocity of the vehicle to which the gyro is attached, while $\boldsymbol{\omega}_p^{(w)}$, $\boldsymbol{\omega}_n^{(w)}$ and $\boldsymbol{\omega}_s^{(w)}$ are the precession, nutation and spin rates of the gyro relative to the vehicle. The spin rate of the gyro is three or more orders of magnitude greater than any of the other rates. That is, under conditions in which a control moment gyro is designed to operate,

$$\|\boldsymbol{\omega}_s^{(w)}\| \gg \|\boldsymbol{\omega}^{(v)}\| \quad \|\boldsymbol{\omega}_s^{(w)}\| \gg \|\boldsymbol{\omega}_p^{(w)}\| \quad \|\boldsymbol{\omega}_s^{(w)}\| \gg \|\boldsymbol{\omega}_n^{(w)}\|$$

We may therefore accurately express the angular momentum of any high-rpm gyro as

$$\{\mathbf{H}_{G_w}^{(w)}\} = [\mathbf{I}_{G_w}^{(w)}]\{\boldsymbol{\omega}_s^{(w)}\} \quad (10.138)$$

Since the spin axis of a gyro is an axis of symmetry, about which the moment of inertia is $C^{(w)}$, this can be written

$$\mathbf{H}_{G_w}^{(w)} = C^{(w)}\boldsymbol{\omega}_s^{(w)}\hat{\mathbf{n}}_s^{(w)}$$

where $\hat{\mathbf{n}}_s^{(w)}$ is the unit vector along the spin axis, as illustrated in Figure 10.25. Relative to the body frame axes of the spacecraft, the components of $\hat{\mathbf{n}}_s^{(w)}$ appear as follows,

$$\hat{\mathbf{n}}_s^{(w)} = \sin \theta \cos \phi \hat{\mathbf{i}} + \sin \theta \sin \phi \hat{\mathbf{j}} + \cos \theta \hat{\mathbf{k}} \quad (10.139)$$

If we let

$$H^{(w)} = C^{(w)}\omega_s^{(w)}$$

then Equation 10.138 becomes, simply,

$$\mathbf{H}_{G_w}^{(w)} = H^{(w)}\hat{\mathbf{n}}_s^{(w)} \quad (10.140)$$

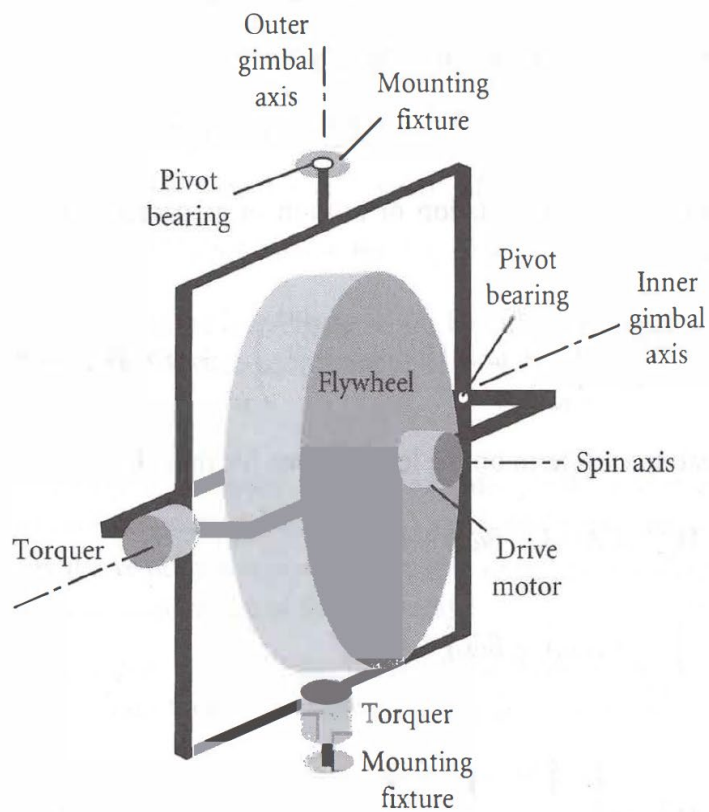


Figure 10.24 Two-gimbal control moment gyro.

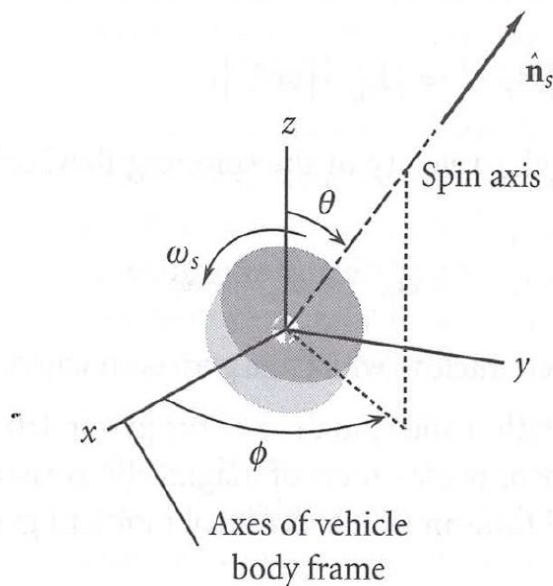


Figure 10.25 Inclination angles of the spin vector of a gyro.

Let us consider the equation of motion of a spacecraft with a single gyro. From Equation 10.136,

$$\left. \frac{d\mathbf{H}_G^{(v)}}{dt} \right)_{\text{rel}} + \boldsymbol{\omega} \times \mathbf{H}_G^{(v)} + \left. \frac{d\mathbf{H}_{G_w}^{(w)}}{dt} \right)_{\text{rel}} + \boldsymbol{\omega} \times \mathbf{H}_{G_w}^{(w)} = \mathbf{M}_{G_{\text{net}}} \quad (10.141)$$

Calculating each term on the left, we have, for the vehicle,

$$\begin{aligned} \mathbf{H}_G^{(v)} &= A\omega_x \hat{\mathbf{i}} + B\omega_y \hat{\mathbf{j}} + C\omega_z \hat{\mathbf{k}} \\ \left. \frac{d\mathbf{H}_G^{(v)}}{dt} \right)_{\text{rel}} &= A\dot{\omega}_x \hat{\mathbf{i}} + B\dot{\omega}_y \hat{\mathbf{j}} + C\dot{\omega}_z \hat{\mathbf{k}} \end{aligned} \quad (10.142)$$

$$\boldsymbol{\omega} \times \mathbf{H}_G^{(v)} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \omega_x & \omega_y & \omega_z \\ A\omega_x & B\omega_y & C\omega_z \end{vmatrix} = (C - B)\omega_y \omega_z \hat{\mathbf{i}} + (A - C)\omega_x \omega_z \hat{\mathbf{j}} + (B - A)\omega_x \omega_y \hat{\mathbf{k}} \quad (10.143)$$

For the gyro,

$$\begin{aligned} \mathbf{H}_{G_w}^{(w)} &= H^{(w)} \hat{\mathbf{n}}_s^{(w)} = H^{(w)} \sin \theta \cos \phi \hat{\mathbf{i}} + H^{(w)} \sin \theta \sin \phi \hat{\mathbf{j}} + H^{(w)} \cos \theta \hat{\mathbf{k}} \\ \left. \frac{d\mathbf{H}_{G_w}^{(w)}}{dt} \right)_{\text{rel}} &= (\dot{H}^{(w)} \sin \theta \cos \phi + H^{(w)} \dot{\theta} \cos \theta \cos \phi - H^{(w)} \dot{\phi} \sin \theta \sin \phi) \hat{\mathbf{i}} \\ &\quad + (\dot{H}^{(w)} \sin \theta \sin \phi + H^{(w)} \dot{\theta} \cos \theta \sin \phi + H^{(w)} \dot{\phi} \sin \theta \cos \phi) \hat{\mathbf{j}} \\ &\quad + (\dot{H}^{(w)} \cos \theta - H^{(w)} \dot{\theta} \sin \theta) \hat{\mathbf{k}} \end{aligned} \quad (10.144)$$

$$\begin{aligned} \boldsymbol{\omega} \times \mathbf{H}_{G_w}^{(w)} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \omega_x & \omega_y & \omega_z \\ H^{(w)} \sin \theta \cos \phi & H^{(w)} \sin \theta \sin \phi & H^{(w)} \cos \theta \end{vmatrix} \\ &= (H^{(w)} \omega_y \cos \theta - H^{(w)} \omega_z \sin \phi \sin \theta) \hat{\mathbf{i}} \\ &\quad + (-H^{(w)} \omega_x \cos \theta + H^{(w)} \omega_z \cos \phi \sin \theta) \hat{\mathbf{j}} \\ &\quad + (-H^{(w)} \omega_y \cos \phi \sin \theta + H^{(w)} \omega_x \sin \phi \sin \theta) \hat{\mathbf{k}} \end{aligned} \quad (10.145)$$

Substituting Equations 10.142 through 10.145 into Equation 10.141 yields a vector equation with the following three components

$$\begin{aligned} A\dot{\omega}_x + H^{(w)} \dot{\theta} \cos \phi \cos \theta - H^{(w)} \dot{\phi} \sin \phi \sin \theta + \dot{H}^{(w)} \cos \phi \sin \theta \\ + (H^{(w)} \cos \theta + C\omega_z) \omega_y - (H^{(w)} \sin \phi \sin \theta + B\omega_y) \omega_z = M_{G_{\text{net}x}} \end{aligned} \quad (10.146a)$$

$$\begin{aligned} B\dot{\omega}_y + H^{(w)} \dot{\theta} \sin \phi \cos \theta + H^{(w)} \dot{\phi} \cos \phi \sin \theta + \dot{H}^{(w)} \sin \phi \sin \theta \\ - (H^{(w)} \cos \theta + C\omega_z) \omega_x + (H^{(w)} \cos \phi \sin \theta + A\omega_x) \omega_z = M_{G_{\text{net}y}} \end{aligned} \quad (10.146b)$$

$$\begin{aligned} C\dot{\omega}_z - H^{(w)} \dot{\theta} \sin \theta + \dot{H}^{(w)} \cos \theta - (H^{(w)} \cos \phi \sin \theta + A\omega_x) \omega_y \\ + (H^{(w)} \sin \phi \sin \theta + B\omega_y) \omega_x = M_{G_{\text{net}z}} \end{aligned} \quad (10.146c)$$

Additional gyros are accounted for by adding the components of Equations 10.144 and 10.145 for each additional unit.

EXAMPLE
10.12

A satellite is in torque-free motion ($\mathbf{M}_{G_{\text{net}}} = \mathbf{0}$). A non-gimbaled gyro (momentum wheel) is aligned with the vehicle's x axis and is spinning at the rate ω_s . The spacecraft angular velocity is $\boldsymbol{\omega} = \omega_x \hat{\mathbf{i}}$. If the spin of the gyro is increased at the rate $\dot{\omega}_s$, find the angular acceleration of the spacecraft.

Using Figure 10.25 as a guide, we set $\phi = 0$ and $\theta = 90^\circ$ to align the spin axis with the x axis. Since there is no gimbaling, $\dot{\theta} = \dot{\phi} = 0$. Equations 10.146 then yield

$$\begin{aligned} A\dot{\omega}_x + \dot{H}^{(w)} &= 0 \\ B\dot{\omega}_y &= 0 \\ C\dot{\omega}_z &= 0 \end{aligned}$$

(Example 10.12
continued)

Clearly, the angular velocities around the y and z axes remain zero, whereas,

$$\dot{\omega}_x = -\frac{\dot{H}^{(w)}}{A} = -\frac{C^{(w)}}{A}\dot{\omega}_s$$

Thus, a change in the vehicle's roll rate around the x axis can be initiated by accelerating the momentum wheel in the opposite direction.

EXAMPLE
10.13

A satellite is in torque-free motion. A control moment gyro, spinning at the constant rate ω_s , is gimbaled about the spacecraft y and z axes, with $\phi = 0$ and $\theta = 90^\circ$ (cf. Figure 10.25). The spacecraft angular velocity is $\boldsymbol{\omega} = \omega_z \hat{\mathbf{k}}$. If the spin axis of the gyro, initially along the x direction, is rotated around the y axis at the rate $\dot{\theta}$, what is the resulting angular acceleration of the spacecraft?

Substituting $\omega_x = \omega_y = \dot{H}^{(w)} = \dot{\phi} = 0$ and $\theta = 90^\circ$ into Equations 10.146 gives

$$\begin{aligned} A\dot{\omega}_x &= 0 \\ B\dot{\omega}_y + H^{(w)}(\omega_z + \dot{\phi}) &= 0 \\ C\dot{\omega}_z - H^{(w)}\dot{\theta} &= 0 \end{aligned}$$

where $H^{(w)} = C^{(w)}\omega_s$. Thus, the components of vehicle angular acceleration are

$$\dot{\omega}_x = 0 \quad \dot{\omega}_y = -\frac{C^{(w)}}{B}\omega_s(\omega_z + \dot{\phi}) \quad \dot{\omega}_z = \frac{C^{(w)}}{C}\omega_s\dot{\theta}$$

We see that pitching the gyro at the rate $\dot{\theta}$ around the vehicle y axis alters only ω_z , leaving ω_x unchanged. However, to keep $\omega_y = 0$ clearly requires $\dot{\phi} = -\omega_z$. In other words, for the control moment gyro to control the angular velocity about only one vehicle axis, it must therefore be able to precess around that axis (the z axis in this case). That is why the control moment gyro must have two gimbals.